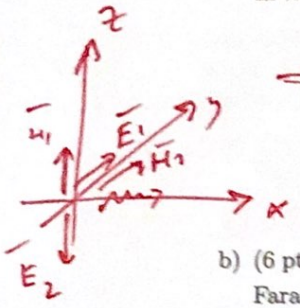
ECE 329 - Sum20, HE3 Name: EK NetID: erhan

1. Three parts of this question are independent:

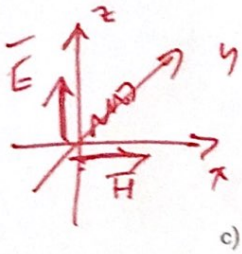
a) (6 pts) Given that $\mathbf{H} = 4\cos(\omega t - \beta x)\hat{z} + 8\sin(\omega t - \beta x)\hat{y}$ A/m, determine the accompanying $\mathbf{E}$ in free space where $\eta = \eta_o = 120\pi\,\Omega$.




$\Rightarrow E = \eta_o \left(4\cos(\omega t - \beta x)\hat{y} - 8\sin(\omega t - \beta x)\hat{z}\right)$ ($\eta_o = 120\pi$)

$= 480\pi\cos(\omega t - \beta x)\hat{y} - 960\pi\sin(\omega t - \beta x)\hat{z}$ V/m.

b) (6 pts) If $\mathbf{E} = \cos(\omega t - \beta y)\,\hat{z}$ V/m in free space so that $\frac{\omega}{\beta} = c$, find the corresponding $\mathbf{B}$ by using Faraday's law $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$. **Hint:** or else use $\mathbf{B} = \mu_o \mathbf{H}$.




$\bar{H} = \frac{1}{\eta_o}\cos(\omega t - \beta y)\hat{x}$

$\bar{B} = \mu_o \bar{H} = \mu_o \sqrt{\frac{\varepsilon_o}{\mu_o}}\cos(\omega t - \beta y)\hat{x} = \frac{1}{c}\cos(\omega t - \beta y)\hat{x}$

$= \frac{1}{3}\times 10^{-8}\cos(\omega t - \beta y)\hat{x}$ Wb/m$^2$.

c) In a homogeneous ideal dielectric with $\epsilon = \epsilon_r \epsilon_o$ and $\mu = \mu_r \mu_o$ plane TEM waves with the following components are observed:

$$\mathbf{E} = \hat{y}u(t + \frac{z}{c/3}) + \hat{x}g(t - \frac{z}{c/3})\,\frac{\mathrm{V}}{\mathrm{m}}$$

and

$$\mathbf{H} = \hat{y}\delta(\frac{15z}{c} - 5t) + \hat{x}\frac{1}{60\pi}u(t + \frac{3z}{c})\,\frac{\mathrm{A}}{\mathrm{m}},$$

where $u(t)$ denotes the unit-step function and $c$ is the speed of light in free space. Using the above information,

i. (2 pts) Determine the intrinsic impedance $\eta$ for the medium.

$\sqrt{\frac{\mu}{\varepsilon}} \rightarrow \eta = 60\pi$

$\rightarrow \eta v_p = \frac{1}{\varepsilon} = 60\pi\,\frac{c}{3} = 60\pi \times 10^8$

ii. (2 pts) Determine the propagation velocity $v$.

$\frac{1}{\sqrt{\mu\varepsilon}} \rightarrow v_p = \frac{c}{3}$

$\rightarrow \frac{\eta}{v_p} = \mu = \frac{60\pi}{c/3} = 60\pi \times 10^{-8}$

iii. (6 pts) Determine $\epsilon_r$ and $\mu_r$.

$\varepsilon_r = \frac{\varepsilon}{\varepsilon_o} = \frac{36\pi \times 10^9}{60\pi \times 10^8} = 6$ $\quad \mu_r = \frac{\mu}{\mu_o} = \frac{60\pi \times 10^{-8}}{4\pi \times 10^{-7}} = 1.5$

iv. (3 pts) Function $g(t)$.

$\frac{g(t)}{\eta} = \delta(-5t) = \frac{\delta(5t)}{5} = \frac{\delta(t)}{5}$

$g(t) = 60\pi\,\delta(-5t)$

$= 12\pi\,\delta(t)$.

$\eta = 60\pi$

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2. On $z = 0$ surface there is a time-varying uniform surface current density

$$\mathbf{J}_s(t) = \hat{y} 4u(t) \sin(\omega t) \frac{\text{A}}{\text{m}}$$

where $u(t)$ is the unit step function and $\omega = 2\pi \times 10^6$ rad/s. This surface current is embedded in free space and radiates a truncated (because of $u(t)$) co-sinusoidal plane TEM wave with $\mathbf{E} = \hat{y} E_y(z,t)$ and $\mathbf{H} = \hat{x} H_x(z,t)$ to both $z > 0$ and $z < 0$ regions. Answer the following questions:

a) (3 pts) What are the numerical values of frequency $\omega$ and wavenumber $\beta$ of the co-sinusoid in the expression for $E_y(z,t)$?

$\omega = 2\pi \times 10^6 \frac{\text{rad}}{\text{s}}$, $\beta = \frac{\omega}{c} = \frac{2\pi \times 10^6}{300 \times 10^6} = \frac{2\pi}{300} \frac{\text{rad}}{\text{m}}$

b) (3 pts) What is the expression for $E_y(z,t)$ in terms of $\omega$, $\beta$, $t$, $z$, and $c$ that is valid for $z > 0$?

$E_y(z,t) = -\frac{\eta_0}{2} J_y\left(t - \frac{z}{c}\right) = -240\pi\, u\left(t - \frac{z}{c}\right) \sin(\omega t - \beta z)$ V/m.

c) (3 pts) What is the average Joule heating rate $\langle \mathbf{J}_s \cdot \mathbf{E} \rangle$ of the radiator at $z = 0$ surface for $t > 0$? Provide a numerical answer.

$\langle \bar{J}_s \cdot \bar{E} \rangle = \langle 4 \sin(\omega t) \left(-240\pi \sin(\omega t)\right) \rangle = -4 \times 240\pi \langle \sin^2(\omega t) \rangle$, $\langle \sin^2 \rangle = 1/2$

$= -480\pi$ W/m$^2$

d) (3 pts) What is the time averaged power per unit area transported by the radiation into the $z > 0$ region for $t > 0$? Provide a numerical answer. **Hint:** related to answer for part (c).

$\langle \bar{E} \times \bar{H} \rangle = \frac{1}{2} \left| \langle \bar{J}_s \cdot \bar{E} \rangle \right| = 240\pi$ W/m$^2$

e) (2 pts) At what instance $t$ will the radiated field first arrive at location $z = 600$ m? Provide the answer in $\mu$s units.

$t = \frac{z}{c} = \frac{600\text{ m}}{300 \frac{\text{m}}{\mu\text{s}}} = 2\ \mu\text{s}$.

f) (2 pts) What is the wavelength $\lambda$ of the co-sinusoid in the expression for $E_y(z,t)$? Provide the answer in units of meter (m).

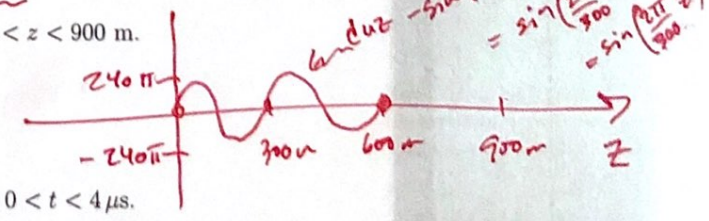
$\beta = \frac{2\pi}{\lambda} = \frac{2\pi}{300} \frac{\text{rad}}{\text{m}} \Rightarrow \lambda = 300$ m.

g) (3 pts) Write down the expression for $E_y(z, t = 2\mu\text{s})$ that is valid for $z > 0$ region.

$E_y(z, 2\,\mu\text{s}) = -240\pi\, u\left(2 - \frac{z}{300}\right) \sin\left(4\pi - \frac{2\pi}{300} z\right)$ V/m

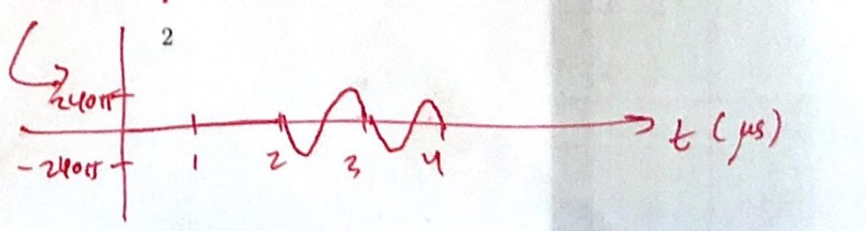
($t$ in $\mu$s, $z$ in m)

h) (3 pts) Plot $E_y(z, t = 2\mu\text{s})$ versus $z$ for $0 < z < 900$ m.




i) (3 pts) Plot $E_y(z = 600\,m, t)$ versus $t$ for $0 < t < 4\,\mu$s.

$E_y(600\text{m}, t) = -240\pi\, u(t-2) \sin(2\pi t - 4\pi)$



3. Consider TEM wave propagation in dry earth — the relevant parameters are $\mu = \mu_o$, $\epsilon = \epsilon_o$, and $\sigma = 10^{-3}$ S/m. Answer the following questions using $\mu_o = 4\pi \times 10^{-7}$ H/m and $\epsilon_o \approx \frac{10^{-9}}{36\pi}$ if and when needed:

a) (7 pts) For the TEM wave wavenumber $\beta$ is much smaller than the damping constant $\alpha$ when the frequency $f \gg 100$ MHz — POSSIBLE or NOT POSSIBLE? Circle one and explain your answer.

Cuz: $\frac{\sigma}{\omega\epsilon} \ll \frac{10^{-3}}{2\pi 10^8 \frac{1}{36\pi\times 10^9}} = \frac{36\times 10^6}{2\times 10^8} = 18\times 10^{-2} = 0.18$ ⇐ Imperfect dielectric

⇒ ∴ $\beta \gg \alpha$ Not the other way around!

b) (7 pts) Frequency $f$ is such that $\sigma \gg \omega\epsilon$ while $\lambda \approx 2\pi\delta$ — POSSIBLE or NOT POSSIBLE? Circle one and explain your answer.

Cuz: $\sigma \gg \omega\epsilon$ ← good conductor ⇒ $\alpha \cong \beta$

∴ $\lambda = \frac{2\pi}{\beta} \cong \frac{2\pi}{\alpha} = 2\pi\delta$

c) (7 pts) If $f = 100$ Hz what is the wavelength $\lambda$?

With $f = 100$ Hz $\frac{\sigma}{\omega\epsilon_0} = \frac{10^{-3}}{2\pi 10^2 \frac{1}{36\pi\times 10^9}} = 18\times 10^4 \gg 1$ ← good cond.

∴ $\alpha = \beta = \sqrt{\pi f \mu \sigma} = \sqrt{\pi\, 10^2\, 4\pi\times 10^{-7}\times 10^{-3}} = 2\pi\times 10^{-4}$

$= \frac{2\pi}{10^4} \Rightarrow \lambda = 10^4$ m

d) (4 pts) If $f = 100$ Hz what is the penetration depth $\delta$.

$\delta = \frac{1}{\alpha} = \frac{10^4}{2\pi}$ m